

Resolver el siguiente sistema, utilizando el método de Gauss-Jordan:

$$\begin{aligned} 6x_1 + 2x_2 + x_3 &= 22 \\ -x_1 + 8x_2 + 2x_3 &= 30 \\ x_1 - x_2 + 6x_3 &= 23 \end{aligned} \quad \left(\begin{array}{ccc|c} 6 & 2 & 1 & 22 \\ -1 & 8 & 2 & 30 \\ 1 & -1 & 6 & 23 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_3 \\ R_3 \rightarrow R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 6 & 23 \\ -1 & 8 & 2 & 30 \\ 6 & 2 & 1 & 22 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_3 \\ R_3 \rightarrow R_1 \end{array} \quad \left(\begin{array}{ccc|c} 1 & -1 & 6 & 23 \\ 0 & 7 & 8 & 53 \\ 0 & 8 & -35 & -116 \end{array} \right) \begin{array}{l} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 - 6R_1 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 6 & 23 \\ 0 & 7 & 8 & 53 \\ 0 & 8 & -35 & -116 \end{array} \right) R_2 \rightarrow \frac{R_2}{7} \quad \left(\begin{array}{ccc|c} 1 & 0 & 5\frac{1}{7} & 24\frac{1}{7} \\ 0 & 1 & \frac{8}{7} & \frac{53}{7} \\ 0 & 0 & -\frac{251}{7} & -\frac{1034}{7} \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 + R_2 \\ R_3 \rightarrow R_3 + 8R_2 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 5\frac{1}{7} & 24\frac{1}{7} \\ 0 & 1 & \frac{8}{7} & \frac{53}{7} \\ 0 & 0 & 1 & 4 \end{array} \right) R_3 \rightarrow -\frac{7}{251} \times R_3 \quad \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 - 5\frac{1}{7}R_3 \\ R_2 \rightarrow R_2 - \frac{8}{7}R_3 \end{array}$$

$$x_1 = 2$$

$$x_2 = 3$$

$$x_3 = 4$$